

Parity-Renormalized Long-Cycle Determinants at a Quadratic Band-Merging Map:

Exact Markov Counts, Noncommuting Small-Noise Limits, and a Logarithmic Cycle Horizon

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Abstract

Fixed-length small-noise localization turns a Gaussian cycle trace into a deterministic periodic-orbit sum, but it gives no uniform control when the cycle length grows. We identify the first exact obstruction at the algebraic band-merging parameter of the quadratic family $f_u(x) = 1 - ux^2$.

Let u_c be the root in $(1, 2)$ of $u^3 - 2u^2 + 2u - 2 = 0$, put $r = u_c - 1$, and set $\lambda = 2u_cr = 1.678573510428\dots$. The three-state Markov graph gives the exact number of distinct real roots of $f_{u_c}^m(x) = x$:

$$N_m = \begin{cases} 1, & m \text{ odd}, \\ 2^{m/2+1} - 1, & m \text{ even}. \end{cases}$$

The subtraction is a genuine boundary-coding multiplicity, not a numerical correction. For every even m we construct a primitive cycle with itinerary $CA(CB)^{m/2-1}$ and prove

$$1 - p_m \sim C_* \lambda^{-m}, \quad C_* > 0.$$

Therefore any uniform use of the interior Gaussian localization theorem can reach at most $m = (\log(1/\sigma))/\log \lambda + O(1)$. This logarithmic cycle horizon is an unconditional consequence of the exact inverse branches.

Write $P_m = \sum_{f_{u_c}^m(x)=x} |1 - (f_{u_c}^m)'(x)|^{-1}$. Odd lengths satisfy the exact identity $P_m = (1 + \lambda^m)^{-1}$. Under a separately stated adapted flat-trace gap hypothesis, $P_m = 1 + (-1)^m + O(\theta^m)$ for some $\theta < 1$. For every fixed $\sigma > 0$, by contrast, the normalized Gaussian operator is strongly positive and compact, its square is trace class, and $\text{tr } \mathcal{K}_\sigma^m \rightarrow 1$. Fixed-length localization then yields the conditional but rigorous noncommutation

$$\lim_{\sigma \downarrow 0} \lim_{k \rightarrow \infty} \text{tr } \mathcal{K}_\sigma^{2k} = 1, \quad \lim_{k \rightarrow \infty} \lim_{\sigma \downarrow 0} \text{tr } \mathcal{K}_\sigma^{2k} = 2.$$

The correct Hilbert–Schmidt determinant removes both the Perron mode and the negative parity mode. At zero noise their exact factor is $1 - z^2$. Dense folded computations through dimension 2560 verify all finite identities. They give $1 + \lambda_-(\sigma) \asymp \sigma^{0.667416}$ on the five smallest resolved noise levels and show a parity lifetime much longer than the logarithmic localization horizon. The exponent $2/3$, the observed even centered-trace rate, and convergence of the negative resonance to -1 are reported as numerical evidence, not as theorems.

Keywords: quadratic map; band merging; periodic-orbit trace; small-noise limit; regularized determinant; parity resonance; Markov partition.

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1 Introduction

A trace formula at one fixed cycle length and a determinant assembled from all cycle lengths are different mathematical statements. For the normalized Gaussian perturbation of a deterministic map, every fixed positive noise produces a compact operator. In the zero-noise limit the kernel concentrates on a graph, and periodic orbits replace smooth spectral data. Even if every fixed trace has a limit, exchanging that limit with an infinite trace series requires uniform information in the cycle length.

The preceding fixed-length result for Gaussian quadratic Markov cocycles proved, under interiority and nondegeneracy of the deterministic cycles, that for each fixed $m \geq 2$,

$$\mathrm{tr} \mathcal{K}_\sigma^m = \sum_{f^m(x)=x} \frac{1}{|1 - (f^m)'(x)|} + O_m(\sigma^2). \quad (1)$$

The proof also isolated boundary contact, multiplier one, and deterioration of the off-orbit action gap as the exact possible sources of nonuniformity [1]. The present paper studies the first of these sources at a parameter where the symbolic geometry is completely explicit.

The parameter is the first algebraic band-merging value of $f_u(x) = 1 - ux^2$. Its invariant core has a period-two decomposition and a three-state Markov matrix with eigenvalues $0, \sqrt{2}, -\sqrt{2}$. The exact geometry and the associated -1 peripheral mode were established in Wang [2]. Here the same partition is used for a different purpose: exact physical periodic-point counts, boundary crowding, long-cycle trace limits, and regularized determinants.

There are two reasons to separate parity before taking any long-cycle limit. First, the deterministic system exchanges two components, so its formal peripheral trace is $1 + (-1)^m$. Second, every fixed positive Gaussian width destroys the exact decomposition: strong positivity leaves only the Perron eigenvalue on the unit circle. Thus the negative mode is long-lived but not permanent. The corresponding limits need not commute.

Periodic-orbit determinants and transfer-operator traces are classical for expanding and piecewise monotone systems [3–6]. The critical map used here is postcritically finite but not uniformly expanding in its original interval coordinate. Correlation decay on its two mixing components follows from standard Misiurewicz-map theory [7–9]; however, an ordinary correlation gap does not by itself identify a flat trace of every power. Rather than silently equating these statements, we place the needed flat-trace estimate in an explicit hypothesis. All Markov counting, boundary crowding, fixed-noise return to the Perron trace, and determinant algebra are proved independently of that hypothesis.

Main results and logical status

The conclusions are deliberately divided into three levels.

- R1. Exact and unconditional.** The physical fixed-point count is 1 at odd length and $2^{m/2+1} - 1$ at even length. At odd length the only point is $r = u_c - 1$, giving $P_m = (1 + \lambda^m)^{-1}$.
- R2. Exact and unconditional.** A primitive even cycle approaches the upper boundary at rate $C_* \lambda^{-m}$. Hence the interiority input in (1) has a logarithmic, not algebraic, uniform cycle horizon.
- R3. Exact and unconditional.** For fixed $\sigma > 0$, $\mathrm{tr} \mathcal{K}_\sigma^m \rightarrow 1$ exponentially in m . This follows from strong positivity, compactness, and the Hilbert–Schmidt eigenvalue bound.
- R4. Conditional theorem.** Under the adapted flat-trace gap [Assumption 4.1](#), $P_{2k} \rightarrow 2$. Together with fixed-length localization, this proves that the two iterated limits are 1 and 2.

R5. Exact determinant algebra. Subtracting $1 + (-1)^m$ from the deterministic coefficients removes the factor $1 - z^2$. At finite noise, subtracting $1 + \lambda_-(\sigma)^m$ removes the actual Perron and parity resonances from the Hilbert–Schmidt regularized determinant.

R6. Numerical evidence. The distinguished negative resonance has a gap consistent with $\sigma^{2/3}$, while the even deterministic centered traces are consistent with $\lambda^{-m/2}$. Neither rate is used in an unconditional theorem.

The result is a long-cycle audit of the Gaussian dynamical construction. It does not identify the negative resonance with a self-adjoint eigenvalue, and it does not promote coefficientwise small-noise convergence to convergence of an unrenormalized infinite determinant.

2 Algebraic band-merging geometry

Let

$$f_u(x) = 1 - ux^2, \quad I = [-1, 1]. \quad (2)$$

Let u_c be the unique root in $(1, 2)$ of

$$u^3 - 2u^2 + 2u - 2 = 0, \quad (3)$$

and define

$$r = u_c - 1, \quad \lambda = 2u_cr. \quad (4)$$

Numerically,

$$u_c = 1.543689012692075\dots, \quad r = 0.543689012692075\dots, \quad \lambda = 1.678573510428318\dots \quad (5)$$

The polynomial in (3) is strictly increasing on $(1, 2)$ and changes sign there. Its defining identity gives

$$0 \mapsto 1 \mapsto -r \mapsto r \mapsto r. \quad (6)$$

In particular, $X = [-r, 1] = f_{u_c}(I)$ is forward invariant, and every periodic point in I belongs to X .

Put

$$A = [-r, 0], \quad B = [0, r], \quad C = [r, 1]. \quad (7)$$

Up to common endpoints,

$$f(A) = C, \quad f(B) = C, \quad f(C) = A \cup B, \quad (8)$$

where $f = f_{u_c}$. Thus the transition matrix is

$$M = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \quad \text{spec}(M) = \{0, \sqrt{2}, -\sqrt{2}\}. \quad (9)$$

The graph is bipartite: A, B form one side and C the other.

2.1 Exact symbolic and physical counts

Let

$$N_m = \#\{x \in I : f^m(x) = x\} \quad (10)$$

count distinct real points, without multiplicity.

Theorem 2.1 (Exact physical fixed-point count). *For every $m \geq 1$,*

$$N_m = \begin{cases} 1, & m \text{ odd}, \\ 2^{m/2+1} - 1, & m \text{ even}. \end{cases} \quad (11)$$

All these fixed points are simple.

Proof. The critical orbit (6) is finite and lands on the repelling fixed point r , whose multiplier is $f'(r) = -\lambda$ with $\lambda > 1$. Thus f is a postcritically finite Misiurewicz map. Standard Markov coding supplies one closed symbolic path for each periodic point not on a partition boundary, and one periodic point for each such path; there are no homtervals, and every periodic orbit is repelling [5, 9]. The only periodic partition endpoint is r : the remaining endpoints enter its orbit by (6).

The number of length- m closed paths with a marked initial state is

$$\text{tr } M^m = \begin{cases} 0, & m \text{ odd}, \\ 2^{m/2+1}, & m \text{ even}, \end{cases} \quad (12)$$

by (9). For even $m = 2k$, the physical fixed point r has the two alternating boundary codes $(BC)^k$ and $(CB)^k$. Every other closed path codes a distinct interior point. Hence one must subtract exactly one from the symbolic count.

For odd m the bipartite graph has no closed path. Nevertheless the exceptional boundary point r is fixed and therefore solves $f^m(x) = x$. There is no other periodic boundary point, so it is the unique solution. Repulsion excludes multiplier one and proves simplicity. $\square$

The difference between (12) and (11) is important. Treating $\text{tr } M^m$ as a physical root count would miss the odd fixed point and double-count the even boundary code.

2.2 Exact odd flat traces

Define the deterministic flat trace

$$P_m = \sum_{f^m(x)=x} \frac{1}{|1 - (f^m)'(x)|}. \quad (13)$$

Corollary 2.2 (Odd trace formula). *For every odd $m \geq 1$,*

$$P_m = \frac{1}{1 + \lambda^m}. \quad (14)$$

In particular, $P_m = O(\lambda^{-m})$ along odd lengths.

Proof. By theorem 2.1, the only point is r . Since $(f^m)'(r) = (-\lambda)^m = -\lambda^m$ for odd m , its weight is $(1 + \lambda^m)^{-1}$. $\square$

3 An explicit boundary cycle and the logarithmic horizon

For $y \in [-r, 1]$, the two inverse branches are

$$g_+(y) = \sqrt{\frac{1-y}{u_c}}, \quad g_-(y) = -\sqrt{\frac{1-y}{u_c}}. \quad (15)$$

The symbols B, C use g_+ and the symbol A uses g_- . Introduce

$$h = g_+ \circ g_+, \quad q = g_+ \circ g_-. \quad (16)$$

At the fixed point r ,

$$h(r) = r, \quad h'(r) = \lambda^{-2}, \quad q(r) = 1, \quad q'(r) = -\frac{1}{2u_c\lambda}. \quad (17)$$

For $m = 2k$, consider the closed word

$$w_{2k} = CA(CB)^{k-1}. \quad (18)$$

Its inverse composition is

$$G_{2k} = q \circ h^{k-1}. \quad (19)$$

Theorem 3.1 (Exponential boundary crowding). *For every $k \geq 1$, the word w_{2k} codes a primitive physical periodic point $p_{2k} \in (r, 1)$. There exists a constant $C_* > 0$ such that*

$$1 - p_{2k} \sim C_* \lambda^{-2k} \quad (k \rightarrow \infty). \quad (20)$$

In particular,

$$1 - p_m = \Theta(\lambda^{-m}) \quad (m \rightarrow \infty, m \text{ even}). \quad (21)$$

Proof. The word (18) is an allowed closed path. Markov coding therefore gives a unique fixed point of (19). It is not an endpoint because the word contains A , and it is primitive because A occurs exactly once: a nontrivial repeated word would contain at least two copies of A .

The map h is increasing on $(r, 1)$, has no fixed point there, and $h(x) < x$ for $x > r$. Its iterates converge to r , uniformly on compact subintervals and, after the first iterate, uniformly on $[r, 1]$. Since $p_{2k} = q(h^{k-1}(p_{2k}))$ and $q(r) = 1$, it follows first that $p_{2k} \rightarrow 1$.

We record the standard attracting-fixed-point estimate. Put $a = h'(r) = \lambda^{-2}$. Linearization of the C^2 map h near r gives a strictly increasing coordinate ψ on its basin such that

$$\psi(h(x)) = a\psi(x), \quad \psi(r) = 0, \quad \psi'(r) = 1. \quad (22)$$

Although h' is singular at the endpoint 1, one iterate sends a neighborhood of 1 into a compact interior interval. Thus $\psi(1) := a^{-1}\psi(h(1))$ is finite and positive, and the same formula extends ψ continuously near 1. Consequently, as $n \rightarrow \infty$, uniformly for x in that neighborhood,

$$h^n(x) - r = a^n \psi(x)(1 + o(1)). \quad (23)$$

Taylor expansion of q at r , followed by (23) with $n = k - 1$, yields

$$\begin{aligned} 1 - p_{2k} &= -q'(r)(h^{k-1}(p_{2k}) - r)(1 + o(1)) \\ &= -q'(r)\psi(1)\lambda^{-2(k-1)}(1 + o(1)) \\ &= C_* \lambda^{-2k}(1 + o(1)), \end{aligned} \quad (24)$$

where $C_* = -q'(r)\psi(1)\lambda^2 > 0$. □

The numerical value approached by $(1 - p_m)\lambda^m$ is $0.4608\dots$, but no numerical value of C_* is needed below.

3.1 Consequence for small-noise localization

Let Δ_m be the smallest distance to $\partial I = \{-1, 1\}$ attained by any point on any cycle satisfying $f^m(x) = x$. The cycle in theorem 3.1 gives

$$\Delta_{2k} \leq 1 - p_{2k} \leq C \lambda^{-2k} \quad (25)$$

for all sufficiently large k .

The fixed-length proof recalled in (1) replaces a truncated row normalizer by a full Gaussian normalizer inside each orbit tube. For a cycle of clearance Δ_m , that step requires at least $\sigma \ll \Delta_m$; its tail error contains $\exp[-\Delta_m^2/(C\sigma^2)]$.

Corollary 3.2 (Logarithmic localization horizon). *Suppose a family of cycle lengths $m = m(\sigma)$ includes every deterministic cycle of that length and is required to satisfy a uniform interior condition $\Delta_m \geq c\sigma$. Necessarily,*

$$m(\sigma) \leq \frac{\log(1/\sigma)}{\log \lambda} + O(1). \quad (26)$$

If the stronger asymptotic condition $\Delta_m/\sigma \rightarrow \infty$ is required, then necessarily

$$\log(1/\sigma) - m(\sigma) \log \lambda \rightarrow +\infty. \quad (27)$$

Proof. Combine the lower requirement on Δ_m with the upper bound (25), take logarithms, and absorb fixed constants into the $O(1)$ term. The same calculation with $\Delta_m/\sigma \rightarrow \infty$ gives (27). $\square$

Remark 3.3 (What the horizon does and does not prove). The corollary is a rigorous obstruction to applying the existing local Gaussian proof uniformly beyond logarithmic length. It does not prove that the actual noisy trace must fail there; a different boundary-layer analysis could in principle continue farther. Such an analysis would have to replace, not merely reuse, the full-Gaussian orbit formula.

4 Deterministic parity centering

The Markov graph and the component exchange suggest the baseline

$$b_m = 1 + (-1)^m. \quad (28)$$

Define centered deterministic coefficients

$$c_m = P_m - b_m. \quad (29)$$

By [corollary 2.2](#), $c_m = (1 + \lambda^m)^{-1}$ for odd m . The even asymptotic requires more than topological counting.

Assumption 4.1 (Adapted flat-trace gap). There are constants $C_0 > 0$ and $0 < \theta < 1$ such that

$$|P_m - 1 - (-1)^m| \leq C_0 \theta^m \quad (m \geq 1). \quad (30)$$

This is intentionally stated at the exact strength used below. A standard route would construct an adapted tower or weighted inverse-branch space on the two mixing components of f^2 , realize P_m as a flat trace, and prove that the remainder after the two peripheral projectors is flat-trace summable. Piecewise-monotone zeta and transfer-operator theory provides the relevant framework [4–6]. The ordinary component correlation gap established for this Misiurewicz map does not, by itself, verify every part of that construction; hence [Assumption 4.1](#) is not declared unconditional here.

Proposition 4.2 (Conditional deterministic long-cycle limit). *Under [Assumption 4.1](#),*

$$P_{2k} = 2 + O(\theta^{2k}), \quad P_{2k+1} = O(\lambda^{-(2k+1)}). \quad (31)$$

In particular, $P_{2k} \rightarrow 2$ and $P_{2k+1} \rightarrow 0$.

Proof. The even assertion is (30); the odd assertion follows unconditionally from (14). $\square$

The direct computations in [section 7](#) give $P_{28} = 1.9985818867$ and an even centered decay consistent with $\lambda^{-m/2}$. Those values support [Assumption 4.1](#) but do not replace its functional-analytic proof.

5 Fixed positive noise and noncommuting limits

For $\sigma > 0$, define

$$w_\sigma(x, y) = \exp \left[-\frac{(y - f(x))^2}{2\sigma^2} \right], \quad (32)$$

$$Z_\sigma(x) = \int_I w_\sigma(x, z) dz, \quad (33)$$

$$q_\sigma(x, y) = \frac{w_\sigma(x, y)}{Z_\sigma(x)}, \quad (34)$$

$$(\mathcal{K}_\sigma g)(x) = \int_I q_\sigma(x, y) g(y) dy. \quad (35)$$

The kernel is continuous and strictly positive on I^2 , and $\mathcal{K}_\sigma \mathbf{1} = \mathbf{1}$.

Theorem 5.1 (Fixed-noise long-cycle trace). *For every fixed $\sigma > 0$, $\mathcal{K}_\sigma$ is Hilbert–Schmidt on $L^2(I)$, $\mathcal{K}_\sigma^2$ is trace class, and*

$$\mathrm{tr} \mathcal{K}_\sigma^m \longrightarrow 1 \quad (m \rightarrow \infty, m \geq 2). \quad (36)$$

More precisely, if $r_\sigma < 1$ is the spectral radius after removing the simple eigenvalue 1, then

$$|\mathrm{tr} \mathcal{K}_\sigma^m - 1| \leq r_\sigma^{m-2} \sum_{j \geq 2} |\lambda_j(\sigma)|^2, \quad (37)$$

where the non-Perron eigenvalues are repeated by algebraic multiplicity.

Proof. Continuity of the kernel on a compact square makes $\mathcal{K}_\sigma$ Hilbert–Schmidt. Therefore its square is trace class [10, 11]. On $C(I)$ the operator has norm one, preserves $\mathbf{1}$, and is strongly positive. Compact Perron–Frobenius theory therefore implies that 1 is algebraically simple and that every other spectral value has modulus strictly less than one. Every nonzero L^2 eigenfunction is continuous because it equals a nonzero scalar multiple of its image under the continuous-kernel operator. Expanding $(\mathcal{K}_\sigma - \lambda)^p g = 0$ shows in the same way that every generalized eigenvector for $\lambda \neq 0$ is a finite linear combination of continuous images $\mathcal{K}_\sigma^j g$. Hence the nonzero spectra on $C(I)$ and $L^2(I)$, including algebraic multiplicities, agree. Compactness then gives a common bound $r_\sigma < 1$ for the non-Perron spectrum.

The eigenvalues of a Hilbert–Schmidt operator are square summable. Lidskii’s theorem applied to $\mathcal{K}_\sigma^m$, $m \geq 2$, gives

$$\mathrm{tr} \mathcal{K}_\sigma^m = 1 + \sum_{j \geq 2} \lambda_j(\sigma)^m. \quad (38)$$

Taking absolute values and extracting r_σ^{m-2} proves (37) and the limit. $\square$

For every fixed $m \geq 2$, all deterministic periodic points are interior to I and repelling. The fixed-length localization theorem (1) therefore applies and gives

$$\lim_{\sigma \downarrow 0} \mathrm{tr} \mathcal{K}_\sigma^m = P_m. \quad (39)$$

Theorem 5.2 (Noncommuting small-noise and long-cycle limits). *The first iterated even-cycle limit is unconditional:*

$$\lim_{\sigma \downarrow 0} \lim_{k \rightarrow \infty} \mathrm{tr} \mathcal{K}_\sigma^{2k} = 1. \quad (40)$$

Under Assumption 4.1, the opposite iterated limit exists and is

$$\lim_{k \rightarrow \infty} \lim_{\sigma \downarrow 0} \mathrm{tr} \mathcal{K}_\sigma^{2k} = 2. \quad (41)$$

Thus, under that explicit deterministic input, the limits do not commute.

Proof. Equation (40) follows immediately from [theorem 5.1](#). For the opposite order, use [\(39\)](#) first and then [proposition 4.2](#). $\square$

The theorem isolates the singularity. At fixed noise the negative parity mode lies strictly inside the unit disk and eventually decays. At zero noise the two components exchange exactly, so the parity contribution survives at every even time. No finite collection of fixed- m limits can decide which behavior controls a simultaneous limit.

6 Parity-renormalized Hilbert–Schmidt determinants

Because $\mathcal{K}_\sigma$ is Hilbert–Schmidt but is not assumed trace class, the natural determinant is the second regularized determinant [\[11\]](#):

$$\det_2(\text{Id} - z\mathcal{K}_\sigma) = \prod_j (1 - z\lambda_j) e^{z\lambda_j} = \exp \left[- \sum_{m \geq 2} \frac{z^m}{m} \text{tr } \mathcal{K}_\sigma^m \right] \quad (42)$$

in its trace-series disk, followed by analytic continuation of the entire product.

6.1 The deterministic parity factor

Under [Assumption 4.1](#), define

$$\hat{D}_{0,\text{bulk},2}(z) = \exp \left[- \sum_{m \geq 2} \frac{c_m z^m}{m} \right]. \quad (43)$$

The series is analytic for $|z| < \theta^{-1} > 1$. The baseline factor is exact:

$$\begin{aligned} \exp \left[- \sum_{m \geq 2} \frac{(1 + (-1)^m) z^m}{m} \right] &= \exp \left[- \sum_{m \geq 1} \frac{z^m + (-z)^m}{m} \right] \\ &= 1 - z^2. \end{aligned} \quad (44)$$

The $m = 1$ term vanishes, so the Hilbert–Schmidt regularization changes nothing in [\(44\)](#). The formal deterministic determinant is therefore

$$\hat{D}_{0,2}(z) = (1 - z^2) \hat{D}_{0,\text{bulk},2}(z). \quad (45)$$

6.2 Finite-noise extraction

Suppose a fixed positive-noise operator has a selected simple negative resonance $\lambda_-(\sigma)$, as in the computations below. Define

$$c_{\sigma,m} = \text{tr } \mathcal{K}_\sigma^m - 1 - \lambda_-(\sigma)^m. \quad (46)$$

Removing the two corresponding factors from [\(42\)](#) gives

$$D_{\sigma,\text{bulk},2}(z) = \exp \left[- \sum_{m \geq 2} \frac{c_{\sigma,m} z^m}{m} \right] \quad (47)$$

$$= \prod_{\lambda_j \notin \{1, \lambda_-(\sigma)\}} (1 - z\lambda_j) e^{z\lambda_j}, \quad (48)$$

and the exact factorization

$$\det_2(\text{Id} - z\mathcal{K}_\sigma) = (1 - z) e^z (1 - z\lambda_-(\sigma)) e^{z\lambda_-(\sigma)} D_{\sigma,\text{bulk},2}(z). \quad (49)$$

Proposition 6.1 (Coefficientwise conditional bridge). *If $\lambda_-(\sigma) \rightarrow -1$ as $\sigma \downarrow 0$, then for every fixed $m \geq 2$,*

$$c_{\sigma,m} \longrightarrow c_m. \quad (50)$$

Consequently every fixed Taylor coefficient of the finite truncations of $D_{\sigma,\text{bulk},2}$ converges to its deterministic centered counterpart.

Proof. Combine (39), $\lambda_-(\sigma)^m \rightarrow (-1)^m$, and the definitions (29) and (46). Every coefficient of fixed degree depends on only finitely many traces. $\square$

The hypothesis $\lambda_-(\sigma) \rightarrow -1$ is strongly supported numerically but is not proved here. More importantly, coefficientwise convergence does not provide a common summable majorant in m . The logarithmic horizon in corollary 3.2 shows why that missing step is substantive.

7 Numerical audit

All calculations use the algebraic root of (3), rather than its printed decimal. Periodic points are generated from every allowed closed Markov word and refined as fixed points of its inverse composition. Duplicate endpoint codes are removed numerically only after the exact count is known. The implementation checks the residual $f^m(x) - x$, the expected root count, and every multiplier.

For finite noise, evenness of f gives an exact nonzero-spectrum folding to $[0, 1]$. On positive midpoint nodes $x_i = (i + 1/2)/n$, the row-normalized matrix has entries proportional to

$$\exp \left[-\frac{(x_j - f(x_i))^2}{2\sigma^2} \right] + \exp \left[-\frac{(-x_j - f(x_i))^2}{2\sigma^2} \right]. \quad (51)$$

The main sweep keeps $n\sigma \simeq 20.48$ and reaches $n = 2048$; a separate $\sigma = 0.01$ resolution audit reaches $n = 2560$. Row stochasticity holds to 4.5×10^{-16} . NumPy, SciPy, and Matplotlib are used for the numerical work [12–14].

7.1 Deterministic counts and traces

Table 1 gives representative exact-word computations. At $m = 28$, all 32767 physical roots are included. The minimum boundary clearance agrees with the distinguished cycle through every enumerated even length.

Table 1: Deterministic long-cycle audit. The centered value is $c_m = P_m - 1 - (-1)^m$. Clearance is $1 - p_m$ for the distinguished even cycle.

m	N_m	P_m	c_m	clearance
1	1	0.3733330432	3.73333×10^{-1}	—
2	3	1.1801429862	-8.19857×10^{-1}	9.89816×10^{-2}
4	7	1.4198783447	-5.80122×10^{-1}	4.27711×10^{-2}
6	15	1.6221410887	-3.77859×10^{-1}	1.70302×10^{-2}
10	63	1.8555077105	-1.44492×10^{-1}	2.41248×10^{-3}
20	2047	1.9887693744	-1.12306×10^{-2}	1.45281×10^{-5}
24	8191	1.9960068858	-3.99311×10^{-3}	1.83678×10^{-6}
28	32767	1.9985818867	-1.41811×10^{-3}	2.31671×10^{-7}

On lengths $12 \leq m \leq 40$, the least-squares slope of $\log(1 - p_m)$ against m is

$$-0.51685054, \quad -\log \lambda = -0.51794433. \quad (52)$$

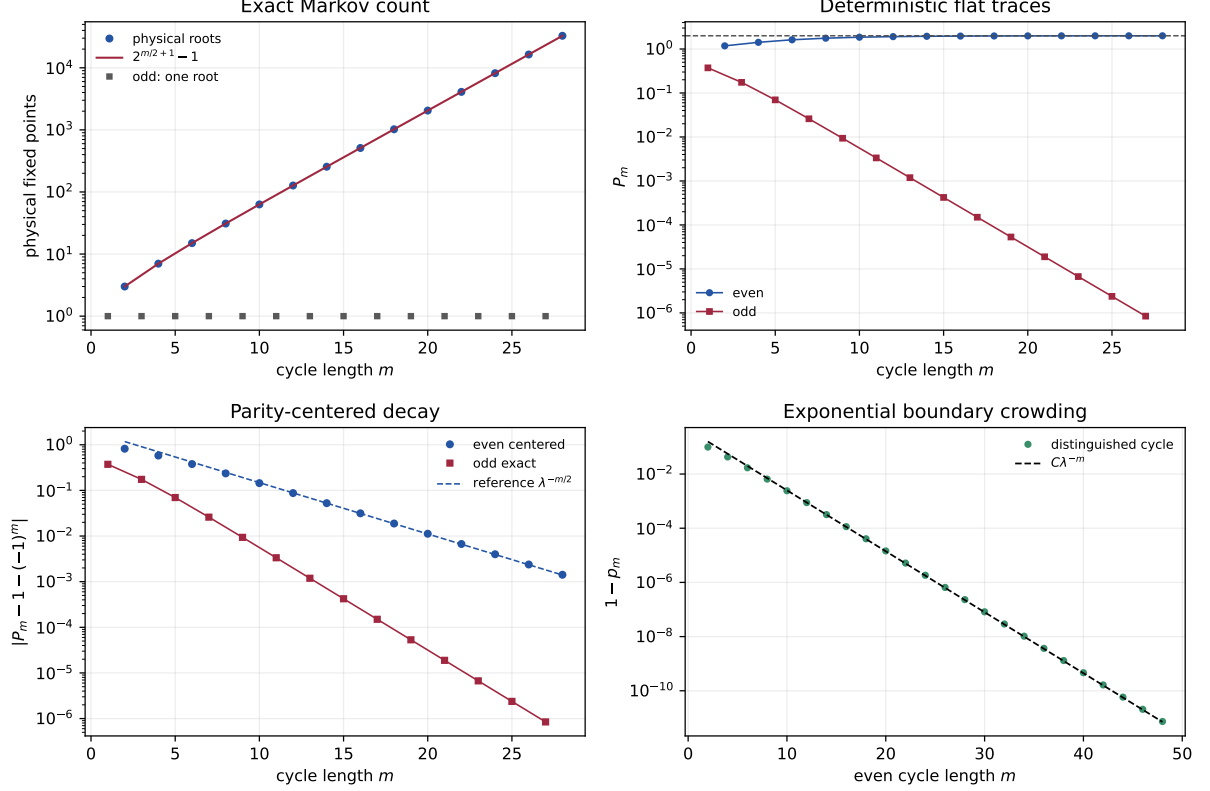


Figure 1: Deterministic audit. Top left: physical root counts lie exactly on $2^{m/2+1} - 1$ at even lengths, while every odd length has one boundary fixed point. Top right: odd traces decay and even traces approach the parity baseline numerically. Bottom left: parity-centered traces; the dashed even rate is a numerical reference, not a theorem. Bottom right: the explicitly constructed boundary cycle follows $C\lambda^{-m}$.

The scaled clearance $(1 - p_m)\lambda^m$ has tail mean 0.460769. This is a numerical check of [theorem 3.1](#), not the source of its proof. For even $14 \leq m \leq 28$, the fitted centered-trace slope is -0.25813959 , close to $-\frac{1}{2} \log \lambda = -0.25897217$.

7.2 Negative resonance and two cycle scales

At every tested noise level the folded matrix has a real negative resonance separated from the remaining spectrum. [Table 2](#) records it along with the bulk spectral radius after removing 1 and λ_- . A log-log fit over the five smallest noise levels gives

$$1 + \lambda_-(\sigma) \simeq 0.2881505 \sigma^{0.6674161}. \quad (53)$$

The scaled gap is nearly constant over the tail. At $\sigma = 0.01$, increasing n from 1024 to 2560 changes λ_- by 2.34×10^{-5} , which is 0.17% of the measured gap.

If (53) persists, the parity e-folding scale is

$$m_{\text{par}}(\sigma) \asymp \frac{1}{1 + \lambda_-(\sigma)} \asymp \sigma^{-2/3}. \quad (54)$$

This is parametrically longer than the rigorous localization ceiling

$$m_{\text{loc}}(\sigma) \lesssim \frac{\log(1/\sigma)}{\log \lambda}. \quad (55)$$

Table 2: Resolved negative parity resonance. The $2/3$ scaling is numerical evidence. The bulk radius removes both the Perron and negative modes.

σ	n	λ_-	$(1 + \lambda_-)/\sigma^{2/3}$	bulk radius
0.080	256	-0.93736537	0.33736	0.46206
0.060	342	-0.95194498	0.31355	0.56193
0.050	410	-0.95883854	0.30328	0.58907
0.040	512	-0.96535455	0.29621	0.57356
0.030	683	-0.97194943	0.29053	0.55714
0.020	1024	-0.97909426	0.28373	0.65643
0.015	1366	-0.98268143	0.28474	0.68226
0.012	1707	-0.98492256	0.28766	0.67713
0.010	2048	-0.98655340	0.28970	0.67467

Table 3: Trace decomposition at $\sigma = 0.015$ and folded dimension 1366. The equality raw = peripheral + bulk holds to rounding.

m	$\text{tr } K_\sigma^m$	$1 + \lambda_-^m$	bulk trace
2	1.18300258	1.96566279	-7.82660×10^{-1}
4	1.48409214	1.93250463	-4.48412×10^{-1}
6	2.21486409	1.90048503	3.14379×10^{-1}
20	1.70543195	1.70510693	3.25018×10^{-4}
100	1.17429098	1.17429098	4.90×10^{-17}
200	1.03037735	1.03037735	1.17×10^{-33}

At $\sigma = 0.01$, the two diagnostics are approximately 74.4 and 8.9. Their separation explains why finite-noise traces can retain a long parity transient even though the termwise deterministic orbit proof has already lost uniform boundary control.

Table 3 illustrates the eventual fixed-noise limit. The bulk part becomes negligible by length 20 at $\sigma = 0.015$; the remaining slow return is almost entirely $1 + \lambda_-^m$.

7.3 Bulk determinant audit

For each finite matrix, (47) truncated at $m = 120$ agrees with the direct eigenvalue product (48) to at worst 2.51×10^{-14} over all tested $z \in [0.25, 1.10]$. This verifies the factorization independently of the deterministic comparison.

The deterministic curve in figure 3 is the explicit 28-term centered trace truncation. At $\sigma = 0.01$, the finite-noise difference is 0.00436 at $z = 0.5$, 0.10322 at $z = 0.9$, and 0.63235 at $z = 1.1$. Approach is clear well inside the unit disk and progressively slower as z weights longer cycles more strongly. This is the expected signature of coefficientwise convergence without a proved common long-cycle bound.

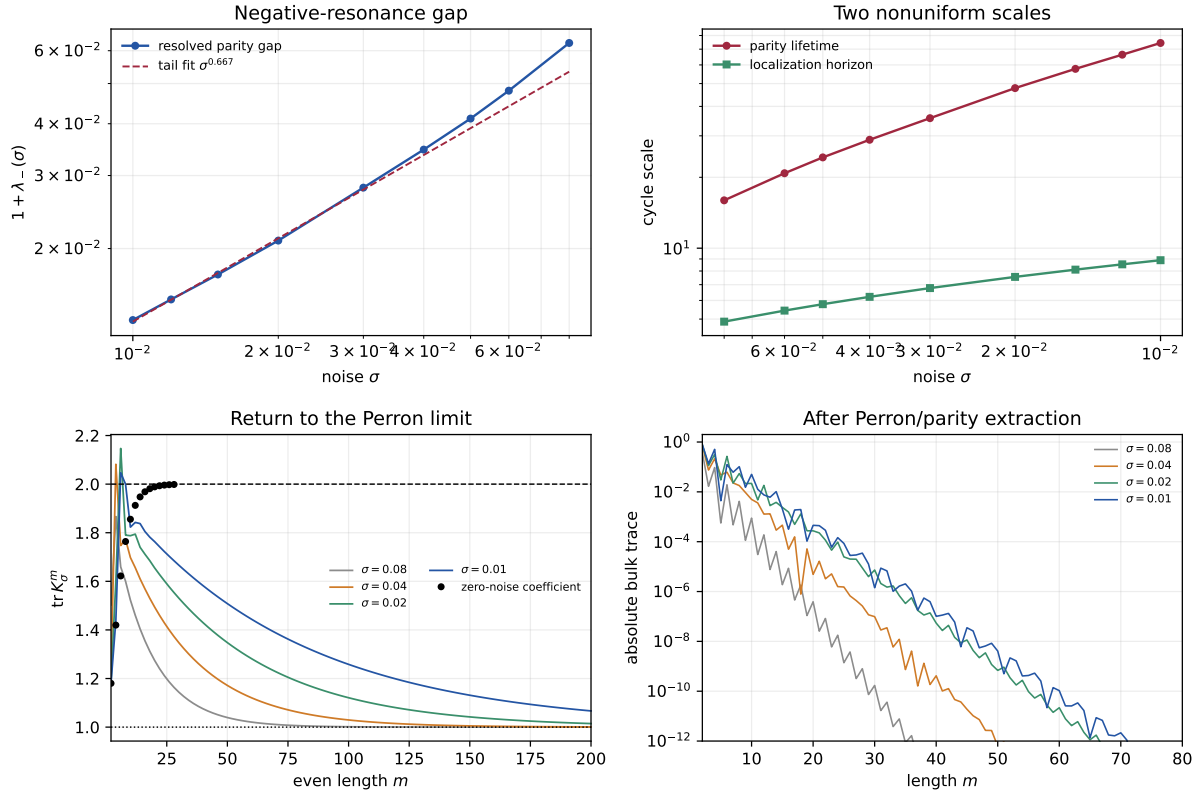


Figure 2: Finite-noise long-cycle behavior. Top left: the negative-resonance gap and its five-point tail fit. Top right: the numerical parity lifetime is already much longer than the rigorous logarithmic localization horizon. Bottom left: every fixed-noise even trace ultimately returns to 1, while the fixed-length zero-noise coefficients approach 2 numerically. Bottom right: after extracting Perron and parity, the remaining traces decay rapidly.

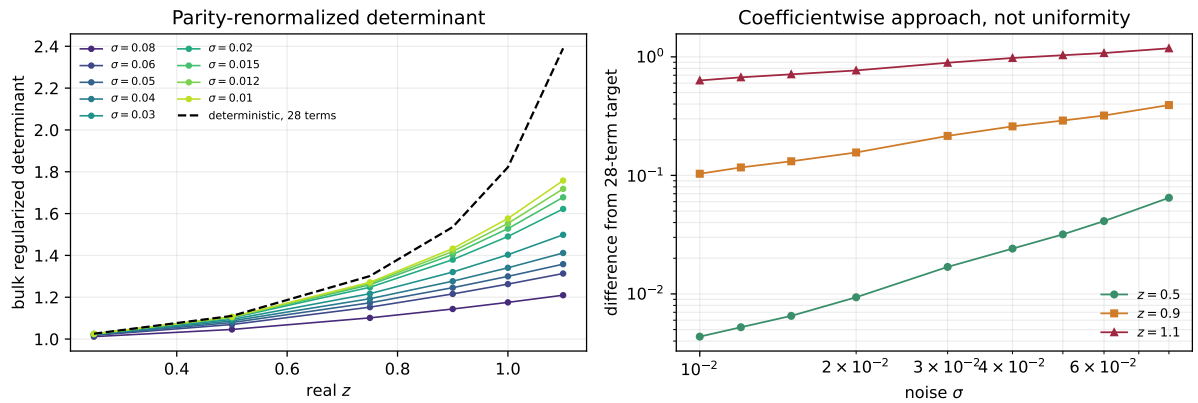


Figure 3: Parity-renormalized regularized determinants. Left: finite-noise bulk products approach the deterministic 28-term centered curve as the noise decreases. Right: the difference is small at $z = 0.5$ but remains substantial at $z = 1.1$, where long coefficients receive greater weight. The comparison is between explicit truncations and is not claimed as uniform determinant convergence.

8 Discussion

The results resolve the first long-cycle question left open by the fixed-length localization theorem.

First, the number of relevant deterministic saddles is known exactly. The factor $2^{m/2}$ comes from the bipartite Markov graph, while the physical -1 correction comes from the double code of r . This distinction is topological and survives any numerical refinement.

Second, an explicit cycle reaches the integration boundary exponentially fast. Therefore a proof based on independent full-Gaussian neighborhoods of every deterministic cycle cannot remain uniform for polynomially long cycles. The natural ceiling is logarithmic in $1/\sigma$. This does not rule out a boundary-layer determinant, but it specifies the new local model that such a construction must contain.

Third, the raw long-cycle trace is the wrong object for comparing positive noise with the deterministic limit. Fixed noise has one permanent peripheral mode; the deterministic map has two. Removing the actual Perron and negative resonances produces a bulk trace whose decay is visible both algebraically and numerically. The factor $1 - z^2$ is not a fitted correction: it follows exactly from the two parity coefficients.

Two analytic tasks remain separate. One is deterministic: construct an adapted flat-trace space and discharge [Assumption 4.1](#). The other is stochastic: prove spectral stability of the negative resonance and derive, or refute, the observed $2/3$ boundary-layer law. Neither task is solved by increasing matrix dimension alone.

9 Conclusion

At the quadratic band-merging parameter, long cycles expose a singular double limit that fixed-length asymptotics cannot see. Exact Markov coding gives the physical periodic-point count and an exact odd trace formula. Explicit inverse branches produce a primitive cycle at distance $\Theta(\lambda^{-m})$ from the boundary, forcing a logarithmic uniform localization horizon. Strong positivity at every fixed noise forces the opposite long-time behavior, namely convergence of the trace to one.

Under one clearly isolated deterministic flat-trace hypothesis, these facts prove that the small-noise and even long-cycle limits are 1 and 2 in opposite orders. The appropriate spectral object is consequently the Perron/parity-renormalized Hilbert–Schmidt determinant. Numerics support a long-lived negative resonance with a $2/3$ gap law, but the manuscript keeps that observation outside the theorem layer. The next theoretical step is therefore well posed: an adapted deterministic flat-trace construction and a small-noise parity boundary-layer analysis, rather than an unqualified raw determinant limit.

Data and code availability

The source code, tests, machine-readable CSV and JSON results, and figure generation script are available in the companion repository [\[15\]](#). The test suite verifies the algebraic orbit, Markov spectrum, exact root counts, periodic residuals, odd trace identity, boundary rate, stochasticity, spectral trace decomposition, and equality of the regularized trace series with the eigenvalue product.

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